Free Rides to FAST

Time limit: 2 seconds | Memory limit: 512 MB

Exams are close, and Bilal has to get from his hostel to the FAST campus in Islamabad using a ride-hailing app. The city is a set of `N` stops numbered `1` to `N`: his hostel is stop `1` and the FAST campus is stop `N`. There are `M` two-way roads between stops, and a ride along road `i` costs w_i rupees, no matter which direction he travels.

The app has just given Bilal `K` free-ride coupons. A coupon can be used on any single ride along one road, making that ride cost `0` rupees. Each coupon can be used only once, and at most one coupon can be applied to a ride. If Bilal travels the same road twice, each ride is a separate ride and needs its own coupon to be free. He does not have to use all of his coupons.

Find the minimum total amount Bilal has to pay to get from stop `1` to stop `N`, or report that the campus cannot be reached.

Input

The first line contains three integers `N`, `M` and `K`: the number of stops, the number of roads and the number of coupons.

Each of the next `M` lines contains three integers `u`, `v` and `w`, meaning there is a two-way road between stops `u` and `v` with ride cost `w`.

Output

Print a single integer: the minimum total cost to travel from stop `1` to stop `N`, or `-1` if it is impossible.

Constraints

- $1 \le N \le 10^5$
- $0 \le M \le 2 \cdot 10^5$
- $0 \le K \le 10$
- $1 \le u, v \le N, u \ne v$
- $1 \le w \le 10^9$
- Several roads may connect the same pair of stops.

Note that the answer may not fit in a 32-bit integer.

Examples

Input	Output
5 6 1 1 2 10 2 4 40 3 5 10 1 4 5 4 5 26 2 3 10	5
4 3 2 1 3 9 1 2 3 2 3 4	-1

Explanation

Example 1. The cheapest ride without coupons is `1 → 2 → 3 → 5` for 30 rupees. With the one coupon, Bilal is better off taking `1 → 4 → 5` (5 + 26 = 31 normally) and using the coupon on the expensive second ride, so he pays only 5 rupees.

Example 2. Stop 4 has no roads, so the campus cannot be reached and the answer is `-1`.